

Excercise - Download climate data

Biologia della Conservazione 25/26

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- ▶ Eurostat
- ▶ NOAA
- ▶ NASA
- ▶ Our World in Data (via GitHub)
- ▶ Copernicus
- ▶ Automated Surface Observing System (ASOS)

- ▶ We will be downloading historical data from the ASOS data repository available at Iowa State.
- ▶ We will working on data from the Italian network of automated weather stations.
- ▶ First lets take a look at the website to see how we can use R to automate the process of downloading the data and eventually analyse them.
- ▶ Automated Surface Observing System (ASOS)

The packages we will be using

```
# Installing the appropriate libraries
install.packages(c("jsonlite", "RCurl", "lubridate",
                  "dplyr", "ggplot2"))
```

```
# Load libraries in the working environment
library(jsonlite)
library(RCurl)
library(lubridate)
library(dplyr)
library(ggplot2)
```


- Now that we know the web site, we can set up a script to automate data downloading

```
# first, we need to specify the date boundaries
# start date in year, month, day format
startDate <- ISOdate(1990, 1, 1)
# end date in year, month, day format
endDate <- ISOdate(2024, 12, 31)

# second, we need to specify the data network parameter
# we want to download the data from
# and the state we want to download the data from
user.network <- c("ASOS")
user.state <- c("IT") #state
```

- We can now start and retrieve the identifier of all the weather stations we are interested in.

```
network <- paste(user.state,  
                 user.network, sep = "__")  
uri <-  
  paste("https://mesonet.agron.iastate.edu/geojson/network/",  
        network, ".geojson", sep="")  
uri
```

```
[1] "https://mesonet.agron.iastate.edu/geojson/network/IT__ASO"
```

- ▶ We use the URL address we have prepared and try to retrieve the information we want.

```
# first we tell R that that we work with an URL
# URL = Uniform Resource Locator
# in this way R will be opening a connection to that URL
# and we will be able to access the website
data <- url(uri)
data
```

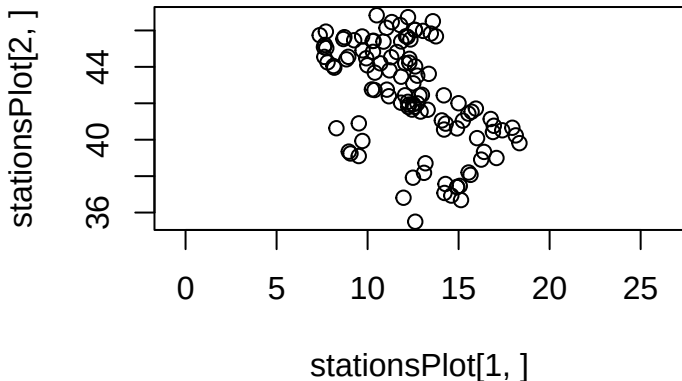
A connection with

```
description "https://mesonet.agron.iastate.edu/geojson/network
class       "url-libcurl"
mode        "r"
text        "text"
opened      "closed"
can read    "yes"
can write   "no"
```

► Now we can use that connection to read the information.

```
jdickt <- fromJSON(data)
# to see what kind of object we have
# you can type and execute jdickt at the prompt
```

```
stationsPlot <-  
  matrix(unlist(jdict$features$geometry$coordinates),  
         nrow = 2 )  
plot(stationsPlot[1,], stationsPlot[2,], asp = 1)
```



- Now that we have all the codes of the weather stations we can work on the URL address where all the specific information of each weather station is stored.

```
service <-  
"https://mesonet.agron.iastate.edu/cgi-bin/request/asos.py?"  
service <- paste(service,  
  "data=all&tz=Etc/UTC&format=comma&latlon=yes&", sep="")  
service <- paste(service, "year1=", year(startDate),  
  "&month1=", month(startDate), "&day1=",  
  mday(startDate), "&", sep="")  
service <- paste(service, "year2=", year(endDate),  
  "&month2=", month(endDate),  
  "&day2=", mday(endDate), "&", sep="")  
it_stations_id <- jdict$features$id
```

- ▶ Set the working directory where to store data.

```
setwd("path/to/your/folder/of/choice")
```

► This is the main for loop to download the all the data.

! Beware!!

The original loop downloads about 4.4GB of data! You should only run it when you have a stable and fast internet connection. Right now we will run a simplified version of the script that should download data from only two stations.


```
it_stations_nm <- jdict$features$properties$sname

for (i in 2:3){
  uri <- paste(service, "station=",
               it_stations_id[i], sep = "")
  print(paste("Network:", network,
             "Downloading:", it_stations_nm[i],
             it_stations_id[i], sep=" "))
  data <- url(uri)
  datestring1 <- format(startDate, "%Y%m%d")
  datestring2 <- format(endDate, "%Y%m%d")
  outfn <- paste(network, "_", it_stations_id[i],
                "_", datestring1, "_to_", datestring2,
                sep="")
  download.file(uri, paste(outfn, ".csv"), "auto")
}
```

 IT__ASOS_EQYS_19900101_to_20201231.csv	Today at 17:33
 IT__ASOS_LIBA_19900101_to_20201231.csv	Today at 17:10
 IT__ASOS_LIBC_19900101_to_20201231.csv	Today at 17:17
 IT__ASOS_LIBD_19900101_to_20201231.csv	Today at 17:12
 IT__ASOS_LIBE_19900101_to_20201231.csv	Today at 17:25
 IT__ASOS_LIBF_19900101_to_20201231.csv	Today at 17:19
 IT__ASOS_LIBG_19900101_to_20201231.csv	Today at 17:22
 IT__ASOS_LIBH_19900101_to_20201231.csv	Today at 17:24
 IT__ASOS_LIBN_19900101_to_20201231.csv	Today at 17:23
 IT__ASOS_LIBP_19900101_to_20201231.csv	Today at 17:28
 IT__ASOS_LIBQ_19900101_to_20201231.csv	Today at 17:33
 IT__ASOS_LIBR_19900101_to_20201231.csv	Today at 17:14
 IT__ASOS_LIBT_19900101_to_20201231.csv	Today at 17:34

Figure 1: This is what my result folder looks like. Note, I have downloaded all 4.4GB of data, but you should only have two files corresponding to “LIMG” and “LIEA”.

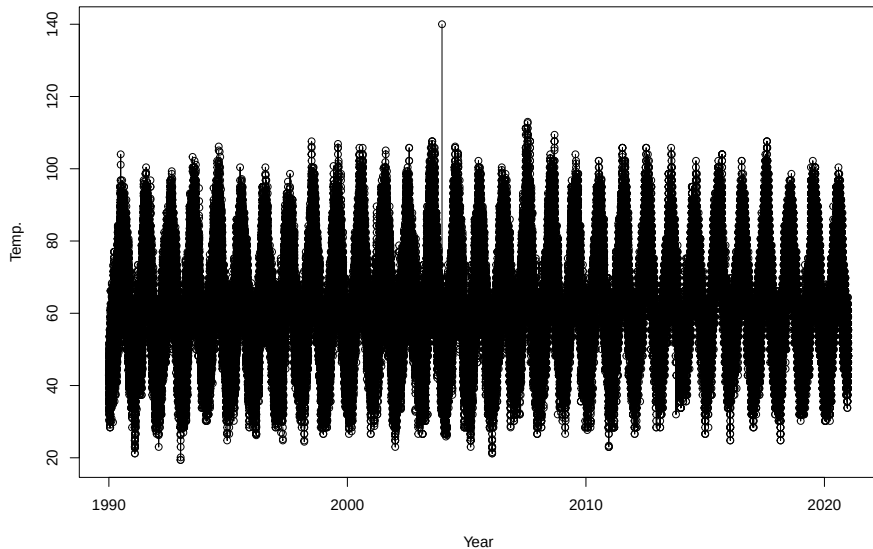
► Time to load the data and look at them.

```
# path_to_csv <- "path/to/your/data"
suffix <- c("LIBA", "LIBC")
files <- list.files(path = path_to_csv,
                    pattern = "*.csv", full.names = T)
files_names <- list.files(path = path_to_csv,
                          pattern = "*.csv", full.names = F)
myfiles <- lapply(files, read.csv, skip = 5,
                  header = T, na.strings = "M")
names(myfiles) <- suffix
myfiles_df <- bind_rows(myfiles, .id = "station_label")
format <- "%Y-%m-%d %H:%M"
myfiles_df$valid <- as.POSIXct(myfiles_df$valid,
                               format = format, tz = "UTC")
myfiles_df[1:13, 2:7]
```

	station		valid	lon	lat	tmpf	dwpf
1	LIBA	1990-01-01	00:00:00	15.7139	41.5381	43.16	41.72
2	LIBA	1990-01-01	01:00:00	15.7139	41.5381	42.80	41.00
3	LIBA	1990-01-01	02:30:00	15.7139	41.5381	42.80	41.00
4	LIBA	1990-01-01	03:00:00	15.7139	41.5381	42.80	41.36
5	LIBA	1990-01-01	04:50:00	15.7139	41.5381	42.80	41.00
6	LIBA	1990-01-01	06:00:00	15.7139	41.5381	43.16	41.00
7	LIBA	1990-01-01	09:00:00	15.7139	41.5381	43.16	41.00
8	LIBA	1990-01-01	12:00:00	15.7139	41.5381	44.60	43.16
9	LIBA	1990-01-01	21:00:00	15.7139	41.5381	44.24	42.08
10	LIBA	1990-01-02	00:00:00	15.7139	41.5381	43.88	42.44
11	LIBA	1990-01-02	03:00:00	15.7139	41.5381	43.16	41.72
12	LIBA	1990-01-02	06:00:00	15.7139	41.5381	43.52	42.08
13	LIBA	1990-01-02	09:00:00	15.7139	41.5381	44.96	43.52

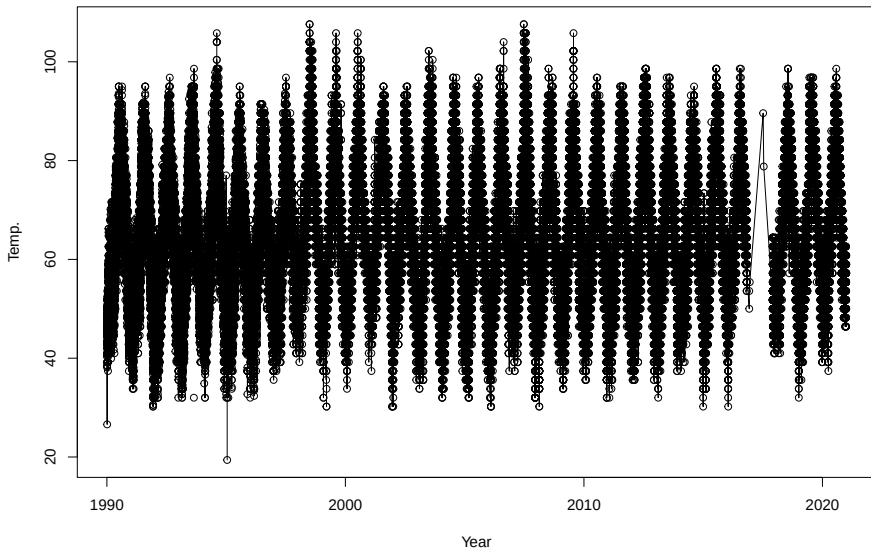
```
plot(date(myfiles_df[myfiles_df$station ==  
                    "LIBA", ]$valid),  
     myfiles_df[myfiles_df$station == "LIBA", ]$tmpf,  
     type = "o",  
     xlab = "Year", ylab = "Temp.",  
     main = "LIBA, Amendola")
```

LIBA, Amendola



```
plot(date(myfiles_df[myfiles_df$station ==  
                    "LIBC", ]$valid),  
     myfiles_df[myfiles_df$station == "LIBC", ]$tmpf,  
     type = "o",  
     xlab = "Year", ylab = "Temp.",  
     main = "LIBC, Crotone")
```

LIBC, Crotone




```
plot(stationsPlot[1,],  
     stationsPlot[2,],  
     asp = 1, pch = 21, bg = "gray")  
points(stationsPlot[1,2:3],  
       stationsPlot[2,2:3],  
       pch = 21, bg = "green")
```

